

ECO387C - Fall 2024

Problem Set 4

Raymond He & Husang Kim

You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

This problem set asks you to implement practical value function iteration *manually*. You will be shown how to and asked to implement value function iteration on your computer in the future.

Question 1 - Manual value function iteration (deterministic)

Consider the deterministic neoclassical growth model with inelastic labor supply from class, with $u(c) = \log(c)$, $\delta = 1$, and $f(k) = k^\alpha$.

1. What is (are) state and control variable(s) for this problem?
2. Write the Bellman equation for this problem.

To solve the Bellman equation, we need to find the value and policy function(s). In this problem however, they are both continuous functions of the state variable, which lives in a domain with infinitely many elements. With finite memory, computers cannot store infinitely many numbers - much less perform non-linear operations on them, so we can only *approximate* the true functions by *discretizing* the state space over a *bounded* interval.

Suppose $\alpha = 0.3$ and $\beta = 0.6$.¹ Discretize the state space (the x is arbitrary) into 5 grid points:

$$x, x' \in \mathcal{X} = \{0.04, 0.08, 0.12, 0.16, 0.20\}.$$

The value function $v()$ is now defined at the 5 points:

$$\{v(0.04), v(0.08), v(0.12), v(0.16), v(0.20)\}.$$

Note that in this discrete state approximation, the standard max operator, which is over a continuous interval of possible x' choices, is actually now only with respect to this coarse grid so it is in fact, the findmax operator.²

¹**Bonus question:** Why may we want to start off with a low value of β ?

²This discrete grid is extremely *coarse*, so the approximation of the value and policy functions will also be poor. The higher the number of grid points, the better the approximation.

3. Referring to the algorithm described below, manually apply two iterations of value function iteration, starting with the initial guess that $v_0(x) = 0 \quad \forall x \in \{0.04, 0.08, 0.12, 0.16, 0.20\}$. Write out all 5 values of $v_2()$ and their associated policy. I.e., fill out the following table:

x	$v_2(x)$	$g_2(x)$
0.04		
0.08		
0.12		
0.16		
0.20		

The value function iteration algorithm is the following:

- Provide an **initial guess** for what the value function is at each point, which in our case will be $v_0(x) = 0 \quad \forall x \in \{0.04, 0.08, 0.12, 0.16, 0.20\}$ ³
- Solve the Bellman equation at each point. I.e. at each point in the state space, find the $x' \in \mathcal{X}$ that attains the maximum.

Hint: The table below shows the right-hand side of the Bellman equation **before** taking the max operator when evaluated at different values of x, x' after the **second iteration**, under the above calibration. The first column has already been filled out for you.

$x \backslash x'$	0.04	0.08	0.12	0.16	0.20
0.04	-1.72				
0.08	-1.49				
0.12	-1.36				
0.16	-1.27				
0.20	-1.20				

- Update the value function evaluated at the 5 grid-points based on step b.
- Compare your initial guess and the updated value. If the values are the same, stop. If they are not, then iteratively repeat Step 2 and 3 until the stopping criterion is met.

³The subscript refers to the iteration.

Question 2 - Manual value function iteration (stochastic)

Consider the stochastic endowment economy discussed in class: Consumers solve

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \log(c_t)$$

subject to

$$c_t + \frac{a_{t+1}}{1+r} = y_t + a_t$$

and a borrowing constraint $a_{t+1} \geq a^{\min}$ where $a^{\min} \leq 0$. The income process (y_t) is stochastic and follows an iid process: $y_t \in \{y^H, y^L\}$, $p(y^H) = p$ and $p(y^L) = 1 - p$

1. What is (are) the state and control variable(s) for this problem?
2. Write the Bellman equation for this problem.

Suppose $\beta = 0.6$, $1 + r = 1.05$, $y^H = 6$, $y^L = 5$, $p = \frac{1}{2}$. Discretize the state space into 5 grid points:

$$a, a' \in \mathcal{A} = \{-1, 0, 1, 2, 3\}.$$

This implies that $a^{\min}$ is effectively set at -1 .

3. Referring to the algorithm described below, manually apply two iterations of value function iteration, starting with the initial guess that $v_0(x) = 0 \quad \forall a \in \{-1, 0, 1, 2, 3\}$. Write out all elements of $v_2(a, y)$ and their associated policy.

Hint: The value function now has two arguments. As shown below, you can think of this as a matrix where rows and columns correspond to a and y , respectively.

Hint: The table below shows the value function when evaluated at different values of (a, y) after the **second iteration** (i.e. $v_2(a, y)$), under the above calibration. The first row has already been filled out for you.

$a \backslash y$	5	6
-1	2.61	2.80
0		
1		
2		
3		

The value function iteration algorithm is the following:

- a Provide an **initial guess** for what the value function is at each point, which in our case will be $v_0(a, y) = 0 \quad \forall a \in \{-1, 0, 1, 2, 3\}, \forall y \in \{5, 6\}$.

- b Solve the Bellman equation at each point. I.e. at each point in the state space, find the $a' \in \mathcal{A}$ that attains the maximum.

Hint: The right-hand side of the Bellman equation now involves taking an expectation of value functions.

Hint: You need to update the value function for each value of y .

- c Update the value function evaluated at the 5 grid-points based on step b.
- d Compare your initial guess and the updated value. If the values are the same, stop. If they are not, then iteratively repeat Step 2 and 3 until the stopping criterion is met.